

Educational Performance and Resource Allocation

Task 4: Power BI Dashboard Development

Objective

The objective of this task is to analyze student performance data using Power BI and create an interactive dashboard that provides meaningful insights into student academic performance.

Dataset Used

Dataset Name: StudentsPerformance.csv

The dataset contains information about:

- Gender
- Parental Level of Education
- Math Score
- Reading Score
- Writing Score
- Lunch Type
- Test Preparation Course

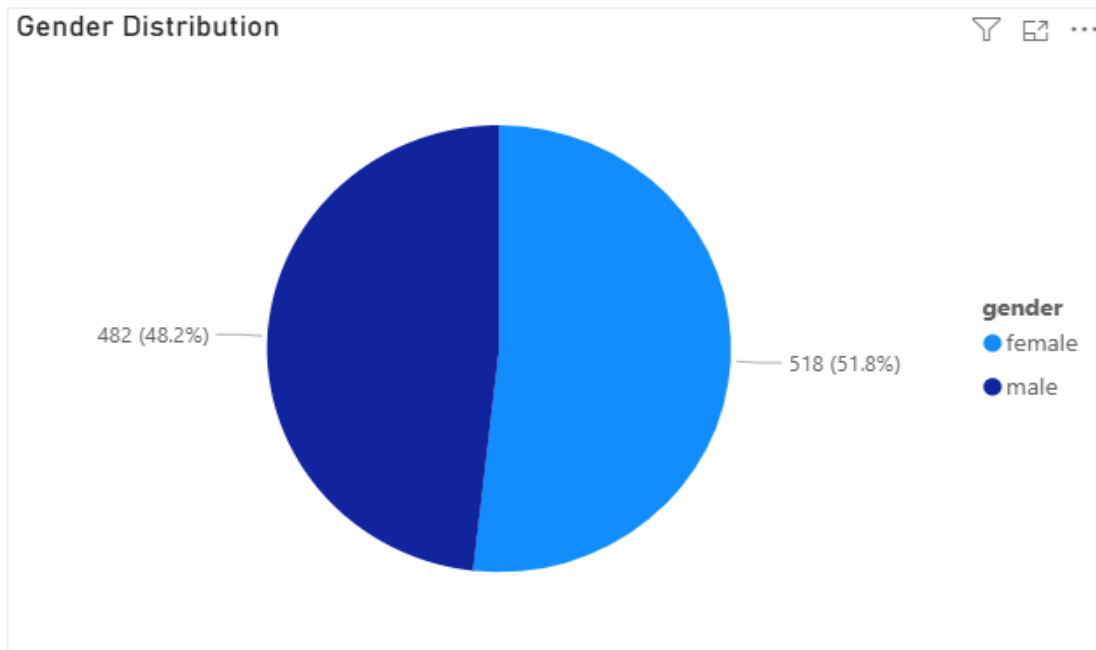
Key Performance Indicators (KPIs)

- Total Students: 1000
- Average Math Score: 66.09
- Average Reading Score: 69.17

Visualizations Created

Gender Distribution (Pie Chart)

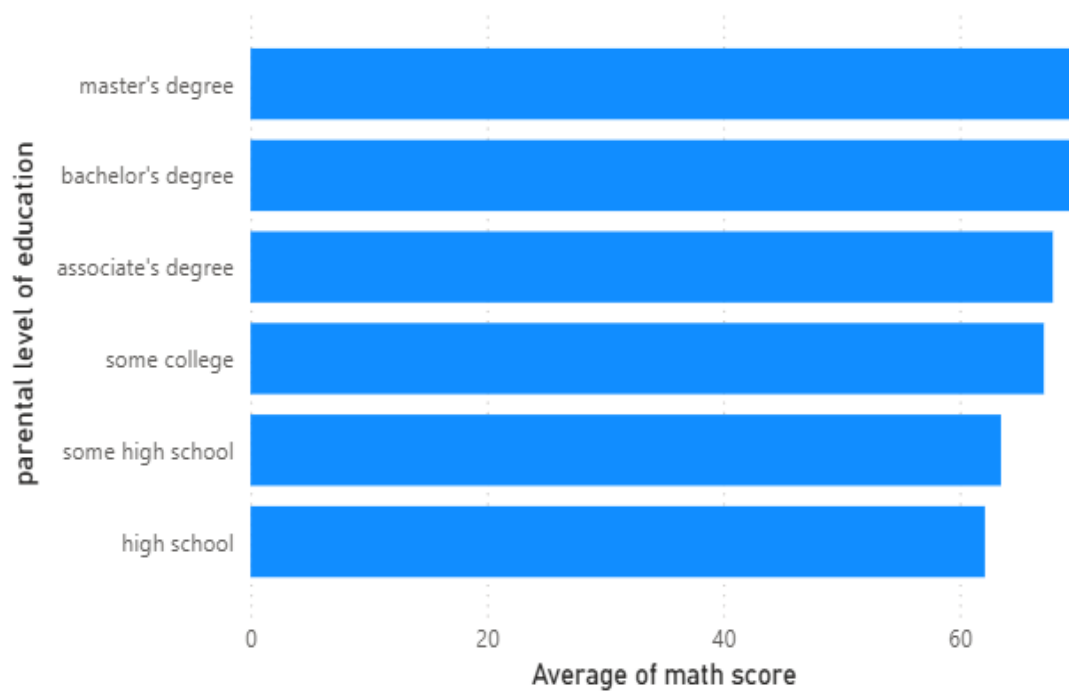
The pie chart represents the distribution of male and female students in the dataset. It helps in understanding the proportion of students based on gender.



Average Math Score by Parent Education (Bar Chart)

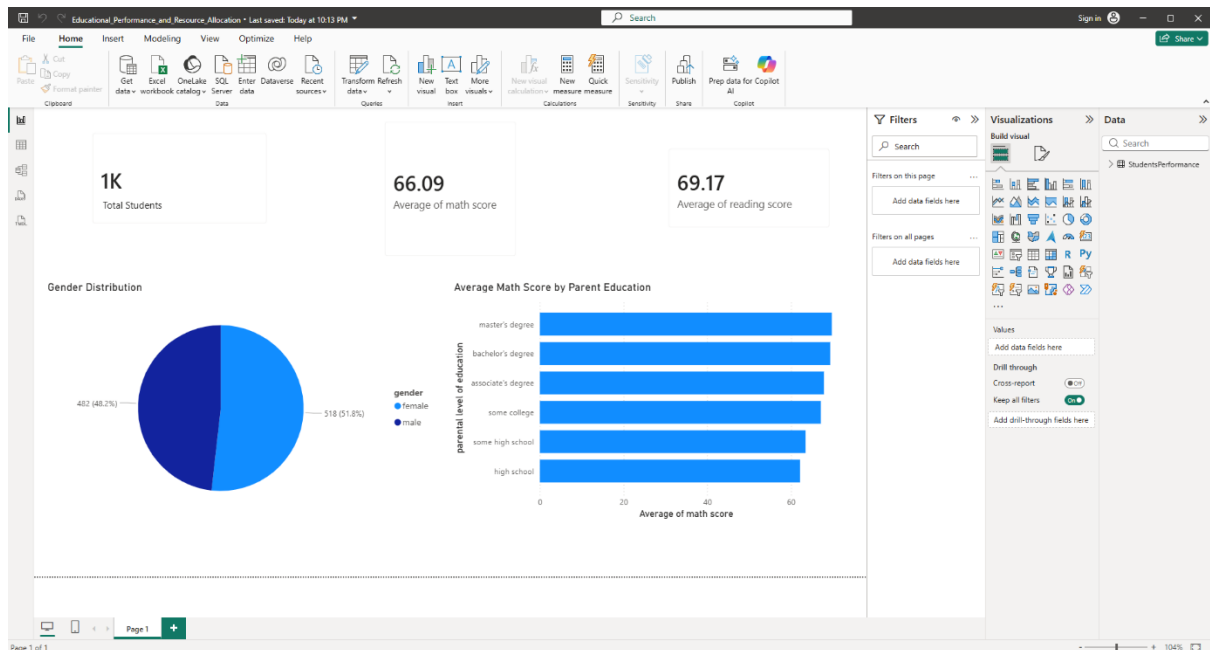
The bar chart shows the average math score achieved by students based on their parents' education levels. This visualization helps identify how parental education influences student academic performance.

Average Math Score by Parent Education



Dashboard Overview

The dashboard combines KPIs and visualizations to provide a comprehensive overview of student performance. It enables quick analysis and comparison of educational factors affecting student scores.



Insights

- Female and male student distribution is nearly balanced.
- Students whose parents have higher education levels generally achieve better math scores.
- The average reading score (69.17) is higher than the average math score (66.09).
- Educational background of parents has a noticeable impact on student academic performance.

Conclusion

The Power BI dashboard successfully analyzes student performance data and presents meaningful insights through interactive visualizations. The dashboard helps understand academic trends, gender distribution, and the influence of parental education on student achievement.